FirstByte 2026: Data Dictionary

This document describes every variable in the three datasets provided with the First Byte 2026 case package. All three datasets cover two full fiscal years of operations, from September 1, 2024 to August 31, 2026. The airline's fiscal year runs from September 1 to August 31. All monetary amounts are in Canadian dollars.

Routes are identified by the code of the departure airport followed by the code of the arrival airport (for example, a flight from Vancouver to Calgary is YVR followed by YYC). Airport codes are listed in the airport table of the case package.

Transactions Dataset

Each row is one customer transaction: a ticket sale, an add-on purchase, a change fee, or a refund.

Variable	Description
`Transaction_ID`	The unique identifier for each transaction.
`Booking_ID`	The identifier for a passenger's trip. A booking can contain several transactions, such as a ticket and a baggage fee. Passengers connecting through a hub have two tickets under the same booking, one for each flight.
`Transaction_Date`	The date and time the transaction was processed. For tickets, this is when the ticket was purchased, not when the flight departed.
`Route_ID`	The route the transaction relates to. For add-ons and refunds, this is the flight they apply to.
`Transaction_Type`	The type of transaction: Ticket, Extra Baggage, Seat Upgrade, Change Fee, Food & Onboard, or Refund.
`Membership_Tier`	The customer's rewards program tier at the time of booking: Non-Member, Blue, Silver, Gold, or Platinum.
`Booking_Channel`	The channel the booking was made through.

Variable	Description
`Payment_Method`	The method the customer used to pay.
`Amount`	The value of the transaction in Canadian dollars. Refunds are recorded as negative amounts.

Costs Dataset

Each row is one category of operating cost for one flight. Every flight appears four times, once for each cost type.

Variable	Description
`Cost_ID`	The unique identifier for each cost record.
`Cost_Date`	The date and time of the flight the cost belongs to.
`Cost_Type`	The category of operating cost: Fuel, Crew, Maintenance, or Food.
`Route_ID`	The route that was flown.
`From`	The departure city of the route.
`To`	The arrival city of the route.
`Amount`	The cost of that type for that flight, in Canadian dollars.

Gates Dataset

Each row pairs one departure gate with one route that departs from it. A gate that serves several routes appears once per route.

Gate leases follow the calendar year rather than the fiscal year. Each lease year runs from January 1 to January 1: 2024-2025 covers January 1, 2024 to January 1, 2025, and 2025-2026 covers January 1, 2025 to January 1, 2026. Records for the 2026-2027 lease year were lost in an accounting error and are not included.

Variable	Description
`City`	The city of the airport where the gate is located. Gate costs apply only to the departure airport.
`Gate_Number`	The gate's identifier within that airport.
`Route_ID`	A route that departs from this gate. Blank if no route is assigned to the gate.
`Annual_Gate_Cost`	The cost of leasing the entire gate for that lease year, in Canadian dollars. This amount is repeated on every row for that gate in that year.
`Year`	The lease year the record belongs to: 2024-2025 or 2025-2026.